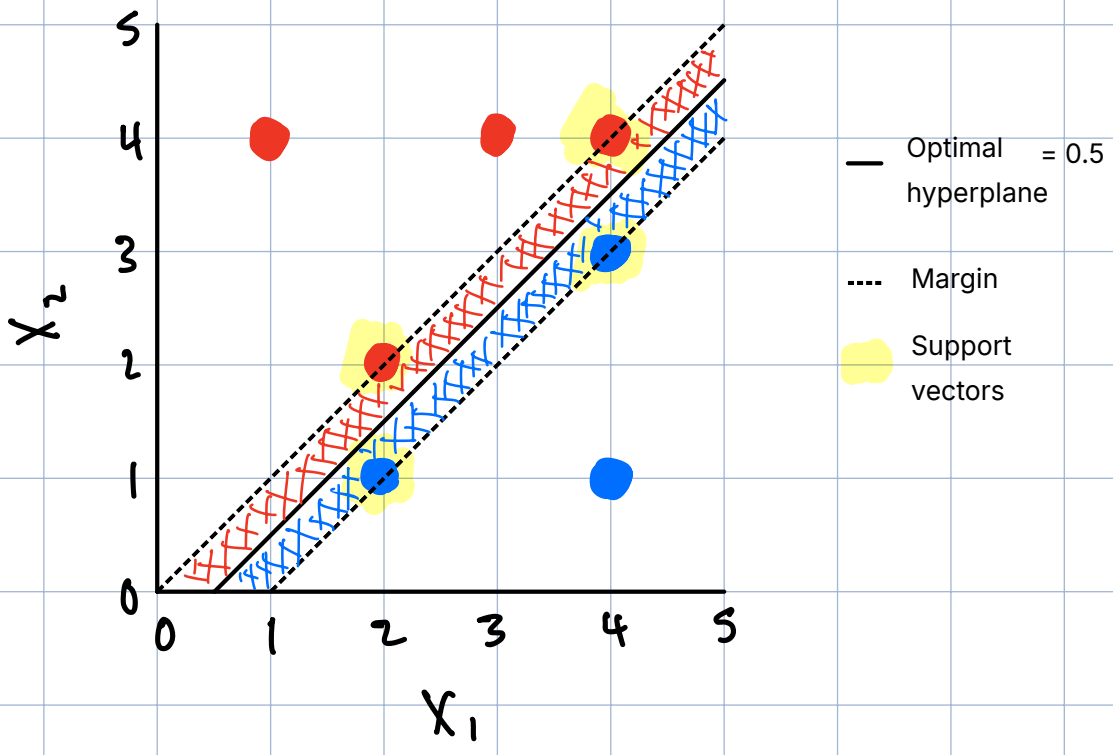
(a), (b), (d), (e)



(b)

$$0.5 - X_1 + X_2 = 0$$

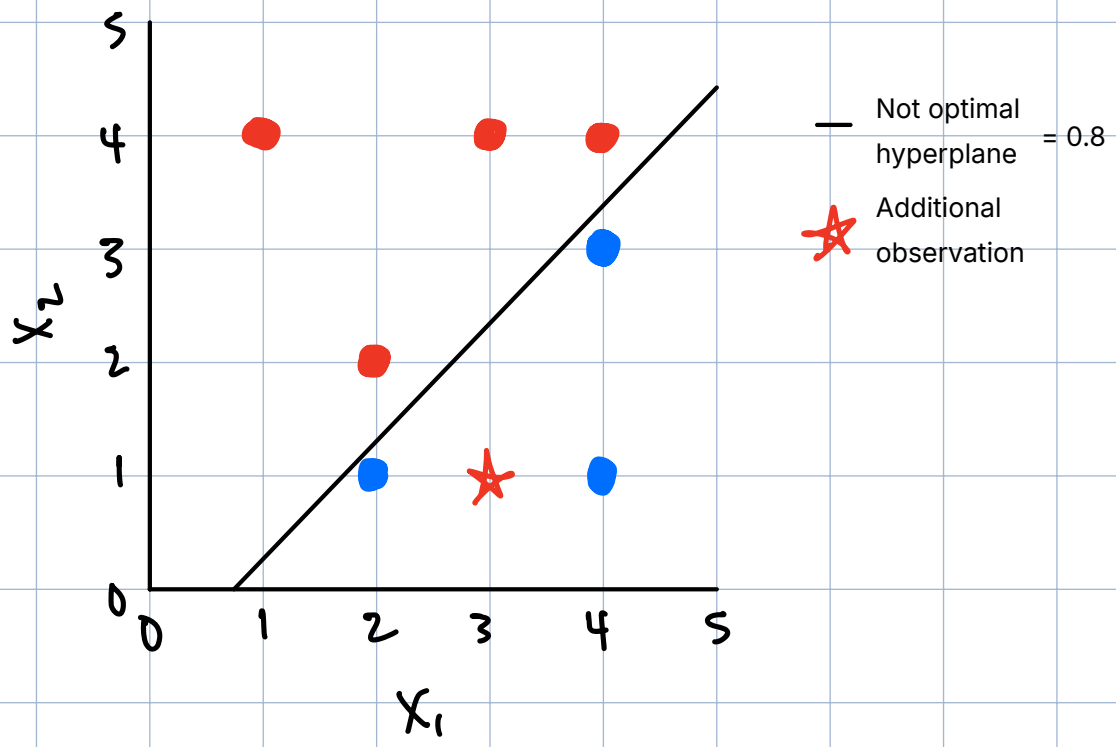
(c)

Classify to Red if $0.5 - X_1 + X_2 > 0$, and classify to Blue otherwise

(f)

It would not be affected because the seventh observation is not a support vector

(g), (h)


5
4
3
2
1
0
0
1
2
3
4
5
X_2
X_1
Not optimal hyperplane
= 0.8
Additional observation

(g)

$$0.8 - X_1 + X_2 = 0$$